

Zainb Maged Arafa Zahran

AI & Full-Stack Developer

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SUMMARY

Full-Stack AI Engineer with strong academic profile currently seeking an AI & Data science master from Queen's university Canda University and hands-on experience shipping production AI applications end to end — from data pipelines and model training to responsive frontends, secure backend APIs, and cloud deployment. Has experience as a Teaching Assistant delivering AI-related coursework, and a Digital Egypt Builders Initiative (DEBI) trainee in AI powered software development. Comfortable across the stack: Python, Django, React, and Scikit-learn, with growing depth in reinforcement learning, NLP, and LLM-powered applications.

EXPERIENCE

Teaching Assistant — Future University in Egypt

Feb 2025 – Jul 2025

- Prepared and delivered tutorials and lab sessions for Machine Learning, Computer Vision, Database Management Systems, Operating Systems, Intelligent Games, Microcontrollers, and Introduction to Data Security.
- Simplified complex AI and systems concepts for undergraduate students and supported them through project-based coursework.

Software Development Trainee — Digital Egypt Builders Initiative (DEBI)

Nov 2024 – Jul 2025

- Completed an intensive software development track covering full-stack engineering and applied AI, culminating in a team-built graduation project.

PROJECTS

LearnYfy — AI-Powered Learning Platform

Feb 2025

- Educational platform that generates personalized study roadmaps and course/book/video recommendations LLM based.
- Stack: Cohere LLM, React, Django, MongoDB. Coordinated the team using Agile methodology.

Course Connect — AI Course Recommendation Engine

Jun 2024

- Built a Random Forest-based recommendation engine for online courses, tuning hyperparameters to reach 95% accuracy.
- Stack: Python, Django, Scikit-learn, HTML, CSS. Led the team and owned the data analysis and model tuning.

AI-Powered Chatbot

2025

- Built a RAG-based chatbot with Cohere's API to answer university-related queries with context-aware, grounded responses.
- Stack: Cohere, RAG, ChromaDB, Python, Flask, React, TypeScript, Tailwind CSS.

Automated Classification of Oral Cancer Lesions

May 2024

- Implemented a Random Forest classifier as a lightweight alternative to a published ResNet-101 approach, matching its ~86% accuracy with substantially shorter training time.

Additional Projects

- Mentorhub (**NBE & AB associate Internship Project**) — multi-role mentorship platform with scheduling, ratings, and an admin dashboard (Python, Django, React, TypeScript, PostgreSQL).
- COVID-19 Patient Recovery Prediction — compared multiple classifiers across evaluation metrics to predict patient recovery.
- Phishing URL detection with CNNs and NLP semantic aware text embeddings.

EDUCATION

Masters in AI & Data Science — Queen's University Canada

Jan 2025 – Aug 2026

B.Sc. in Electrical Engineering — Computer & Systems — Ain Shams University

Sep 2019 – Jul 2024

SKILLS

Machine Learning & Deep Learning: Scikit-learn, TensorFlow, PyTorch, NumPy, Pandas, Matplotlib, model training & evaluation

AI & Data Science: NLP, LLM fine-tuning, RAG-based chatbots, reinforcement learning, exploratory data analysis

Frontend: React, TypeScript, HTML5, CSS, Tailwind CSS, Bootstrap

Backend & Databases: Django, REST APIs, PostgreSQL, MySQL, MongoDB

Cloud & DevOps: AWS (EC2, S3), Docker, Kubernetes, CI/CD pipelines, Terraform, Spark / PySpark

Collaboration: Git, GitHub, GitHub Projects, Agile/Scrum sprint planning

LANGUAGES

English — Fluent (B2, Berlitz) | French — Basic